Cheng, Xinghua

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Research Interests

My research focuses on understanding the nexus between carbon, water, and energy in the face of global change. Guided by research questions that emerge from terrestrial ecosystem predictions and forecasting, I use three main approaches: 1) spatiotemporal data analysis to interpret field and satellite remote sensing observations; 2) hybrid of process-based ecosystem models and AI to understand underlying mechanisms behind ecological processes and implement ecosystem predictions and forecasting across scales; 3) geospatial information technologies to generate novel datasets.

Education

University of Connecticut — Storrs, USA
Ph.D. in Natural Resources and the Environment — 2023-June 2026 (Expected)
Committee: Drs. James Knighton (Chair), Zhe Zhu (co-adviser), Ashley Helton (UConn), Yiqun Xie (UMD), Liyin He (Duke)
Dissertation: Decoding the Impulse of Agroecosystems: Crop Physiology, Process-based Models, and AI

The Hong Kong Polytechnic University — Hung Hom, Hong Kong
M.S. in Geomatics — 2020

Nanjing Normal University — Nanjing, China
B.S. in Geographical Information Science — 2018

Professional Appointment

Cary Institute of Ecosystem Studies — New York, USA
Catskill Science Student Research Fellow — 2025

Southwest Jiaotong University — Chengdu, China
Research Associate — 2022-2023

The Hong Kong Polytechnic University — Hung Hom, Hong Kong
Research Associate — 2020-2022

Publications

In preparation

[1] **Cheng, X.** et al., Towards Ecosystem Productivity and Crop Yields Prediction under Extreme Droughts and Precipitation.

[2] **Cheng, X.** et al., Scalable Agroecosystem Predictions: From Calibration to Parameter Learning.

[3] **Cheng, X.** et al., Field-scale Corn Yield Impacts of Changes in Growing Periods.

First-authored paper

[1] **Cheng, X.**, Tao, B., Qiu, S., Zhu, Z., Ruane C.A., Ren, W. (2025). Long-term Changing Patterns of Field-scale Corn and Soybeans Phenology in the US Midwest. ***Journal of Geophysical Research: Biogeosciences***, 131(1). e2025JG009028.https://doi.org/10.1029/2025JG009028.

[2] **Cheng, X.**, Li, Z. (2025). Can we quantify the relationship between multispectral remote sensing images and land use and land cover maps? An explicit information transfer model based on Boltzmann entropy. ***GIScience & Remote Sensing***, *62(1), 2478689.* https://doi.org/10.1080/15481603.2025.2478689.

[3] **Cheng, X.**, Yan W. (2025). Inferring Radiometric Quality of Multispectral Airborne Laser Scanning Data ***Journal of Geovisualization and Spatial Analysis***, 9(12). https://doi.org/10.1007/s41651-025-00213-8.

[4] **Cheng, X.**, Li, Z. (2024). Modeling Information Flow from Multispectral Remote Sensing Images to Land Use and Land Cover Maps for Understanding Classification Mechanism. ***Geospatial Information Science***, 27(5), 1568-1584. https://doi.org/10.1080/10095020.2023.2275625.

[5] **Cheng, X.**, Li, Z. (2021). Predicting the Lossless Compression Ratio of Remote Sensing Images with Configurational Entropy. ***IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing***.14, 11936-11953. https://doi.org/10.1109/JSTARS.2021.3123650.

[6] **Cheng, X.**, Li, Z. (2021). Configurational Entropy for Optimizing the Encryption of Digital Elevation Model Based on Chaos System and Linear Prediction. ***Applied Sciences***, 11(5), 2402. https://doi.org/10.3390/app11052402.

[7] **Cheng, X.**, Li, Z. (2021). Using Boltzmann Entropy to Measure Scrambling Degree of Grayscale Images. In *Proc. the 2021 IEEE 5th International Conference on Cryptography, Security and Privacy (CSP 2021)*, Zhuhai, China, 181-185. https://doi.org/10.1109/CSP51677.2021.9357492.

[8] **Cheng, X.**, Li Z. (2020). How Does Shannon's source coding theorem Fare in prediction of image compression ratio with current algorithms? In *ISPRS Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*. https://doi.org/10.5194/isprs-archives-XLIII-B3-2020-1313-2020.

Co-authored paper

[9] Huang et al. (including **Cheng, X.**) (2025). Remotely sensed high-resolution soil moisture and evapotranspiration: Bridging the gap between science and society. ***Water Resources Research***, 61(5), e2024WR037929. https://doi.org/10.1029/2024WR037929.

[10] Li, X., Zhao, L., Wang, S., **Cheng, X.**, & Wang, L. (2025). Unstable permafrost regions experience more severe heatwaves in a warming climate. ***npj Climate and Atmospheric Science***, 8(1), 147. https://doi.org/10.1038/s41612-025-01037-5.

[11] Wang, Y., Mao, J., Brelsford, C. M., Ricciuto, D. M., Yuan, F., Shi, X., Rastogi, D., Mayes, M. M, Kao, S., Warrenm, W.J, Griffth, A. N, **Cheng, X.**, Weston, J.D, Zhou, Y., Gu, L., Thornton, P. E. (2024). Thermal, water, and land cover factors led to contrasting urban and rural vegetation resilience to extreme hot months. ***PNAS nexus***, 3(4), 147. https://doi.org/10.1093/pnasnexus/pgae147.

[12] Chen, Y., Liu, Q., Yang, J., **Cheng, X.,** Deng, M. (2024). Spatially constrained statistical approach for determining the optimal number of regions in regionalization. ***International Journal of Geographical Information Science***, 38(10), 2108-2147, https://doi.org/10.1080/13658816.2024.2372779.

[13] Li, S., Hu, G., **Cheng, X.**, Xiong, L., Tang, G. Josef, S. (2022). Integrating topographic knowledge into deep learning for the voids filling of digital elevation models. ***Remote Sensing of Environment***, 260:112818, https://doi.org/10.1016/j.rse.2021.112818.

[14] Zhou, R., Lu, K., Long, Y., Lu, J., **Cheng, X.**, Hu, D., Gu, Y. (2017). A survey on social image understanding. In *Proc. 2017 IEEE International Conference on Behavioral, Economic, Socio-cultural Computing (BESC)*, Krakow, Poland, pp. 1-5. https://doi.org/10.1109/BESC.2017.8256394.

Grant

Current Extramural Grant (~$14, 000)

Assessment of water quality variability in the West of Hudson watersheds influenced by forest phenological shifts under climate change, **lead student PI: Xinghua Cheng**, co-PIs: Drs. James Knighton, Yingying Xie. Catskill Science Collaborative Program Funded by New York State through Environmental Protection Funds, and with support from the NYC Department of Environmental Protection. 5/2025-12/2025. News Release

Research Projects

- **Assessment of water quality variability in the West of Hudson watersheds influenced by forest phenological shifts under climate change**
  - Led the model development for analyzing seasonality of stream nitrate concentration
  - Engaged with resource managers from NYC DEP and weekly reported project progress
  - Prepared a final project report
- **Development of Boltzmann-entropy-based image approaches for remote sensing image analysis**
  - Developed Boltzmann-entropy-based information flow models from scratch
  - Published research articles as the leading author
  - Prepared a final project report

Teaching & Mentorship

The Hong Kong Polytechnic University Hung Hom, Hong Kong

- Course: LSGI3242A Digital Terrain Modeling Fall 2020
  - Teaching the concepts of fundamental GIS and their applications (e.g., data visualization, spatial analysis)
  - Designing the report template (e.g., format, figures) for homework to help students improve writing skills
  - Creating lab tutorial from scratch
  - Designing and grading homework

Journal Review Service

Global Ecology and Biogeography, Remote sensing of environment, Geospatial information science, IEEE transactions on geoscience and remote sensing, IEEE geoscience and remote sensing letters, IEEE journal of selected topics in applied earth observations and remote sensing, Neural Processing Letters, International journal of applied earth observation and geoinformation, Remote sensing letters.

Honors & Awards

2025 Catskill Student Research Fellow
2025 Student Scholarship for the Catskill Environmental Research & Monitoring Conference
2025 Graduate Student Fellowship
2024 Doctoral Student Travel Award
2019 LSGI Scholarship for Outstanding Academic Performance
2018 Outstanding Undergraduate Thesis
2018 Third-class Scholarship for Excellent Student
2017 Second-class Scholarship for Excellent Student
2016 Second Prize in the 5th National GIS Skills Competition
2016 Third-class Scholarship for Excellent Student
2015 Award of Pacemaker to Merit Student

Conference Talks and Posters

2025 **Cheng, X.**, Gelda, R., Mukundan, R., Dangol, S., Xie, Y. "Assessment of Nitrate Variability Influenced by Forest Phenological Shifts in the WOH Watershed under Climate Change". In Catskill Environmental Research & Monitoring (CERM) Conference, New York, USA, October 22-24, 2025. (Poster)

2024 **Cheng, X.**, Tao, B., Qiu, S., Zhu, Z., Ruane C.A., Ren, W. "Mapping Two Decades of Field-scale Corn and Soybean Phenometrics in the US Midwest by Fusing Time Series of Landsat and MODIS Data". In American Geophysical Union Annual Conference, Washington, D.C., USA, December 9-13, 2024. (Poster)

2023 **Cheng, X.**, Li, Z. "Modeling Energy Flow from Landscape Gradients to Mosaics with the Law of Energy Conservation". In COS 264: Modeling, Ecological Society Meeting 2023, Portland, USA, August 6-11, 2023. (Oral)

2021 **Cheng, X.**, Li, Z. "Effects of Upscaling Functions on Calculating Thermodynamic Entropy of Numerical Raster Data: A Case Study on Aggregation". In T-08: Modeling, North American Regional Association of the International Association for Landscape Ecology (IALE-NA) Annual Meeting, Neveda, USA, April 8-16, 2021. (Oral)

Skills

Process-based Models	APSIM, *ecosys*
Programming Languages	Python, R, C, C#, C++, Java, HTML5, JavaScript
Deep Learning Frameworks	PyTorch
Operating Systems	Windows families, MacOS, Linux families
Software and Tools	NCL, WRF, MATLAB, NetCDF, Visual Studio, ArcGIS, ENVI

References

Dr. James Knighton
Assistant Professor (2020-)
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Dr. Yiqun Xie
Assistant Professor (2020-)
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Dr. Liyin He
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